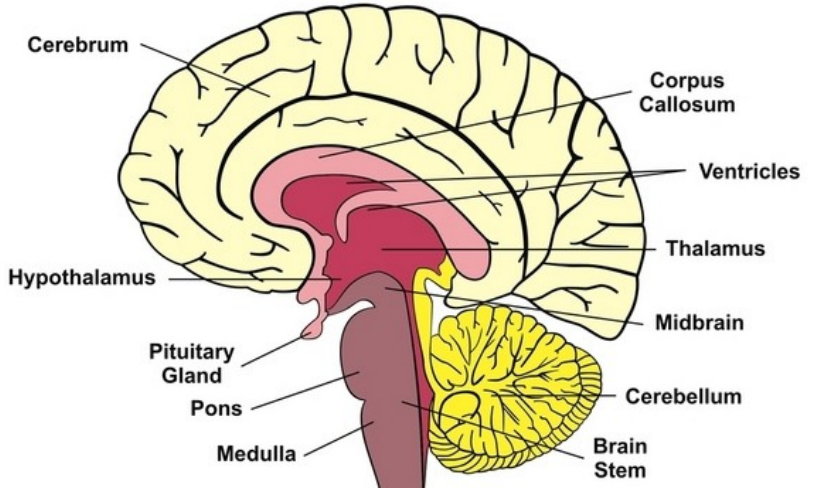
CNS – The Brain



The Nervous System is Highly Organised

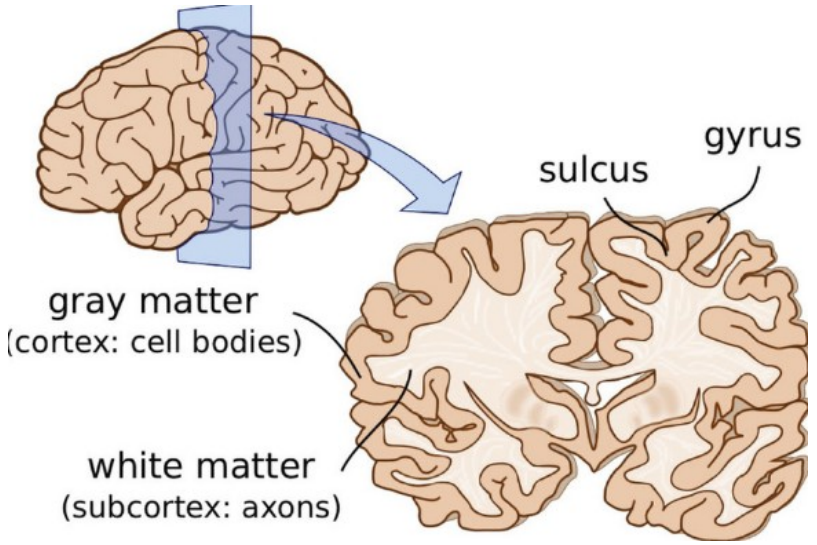
The Brain

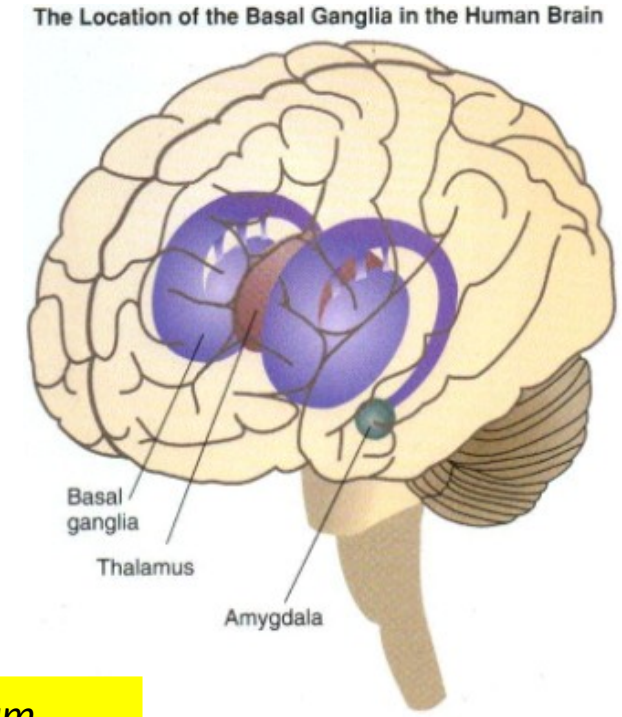
- Complex organ in structure and function
- Still not fully understood
- Works as an integrated whole
- For Human Bio ATAR, we look at major parts that have major functions
- Damage to any part can profoundly affect function.
- We will look at:
  - Cerebrum
  - Cerebellum
  - Hypothalamus
  - Medulla Oblongata
  - Corpus Callosum



The Cerebrum

- Biggest part of the brain
- Cerebral cortex:
  - Outer layer
  - Grey matter (cell bodies)
- White matter
  - Below cortex
  - Axons
- Basal Ganglia
  - Deep within cerebrum
  - Additional grey matter

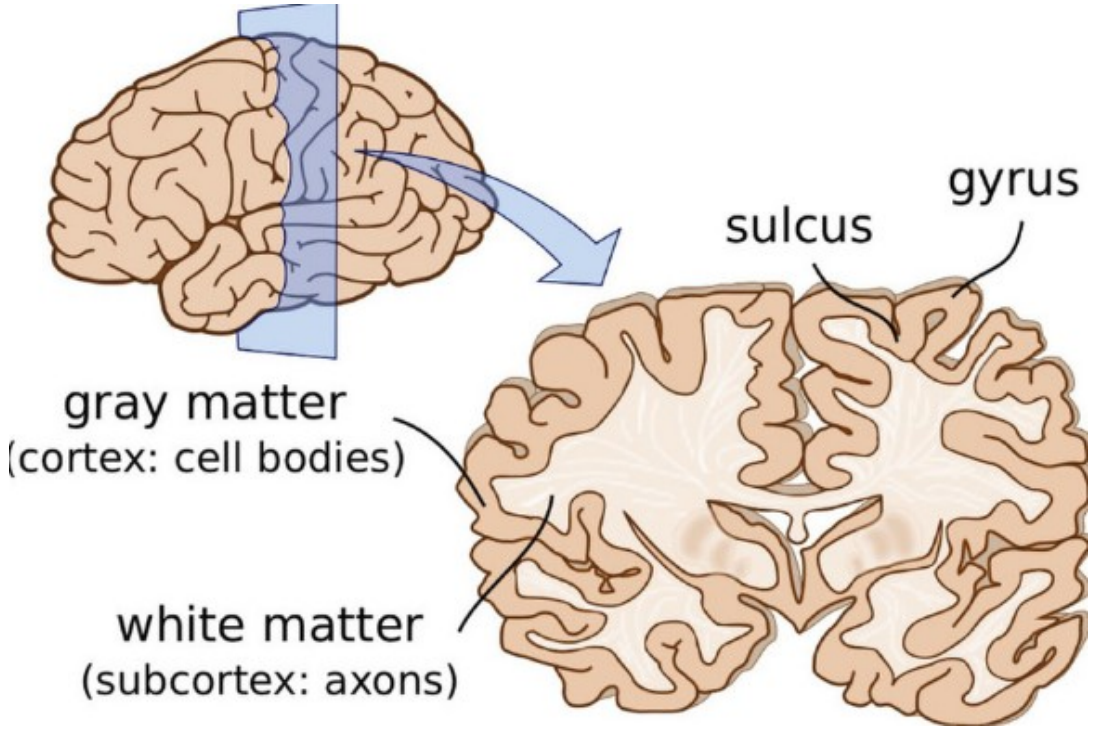




Learning Aim: Describe the location and main functions of the cerebrum

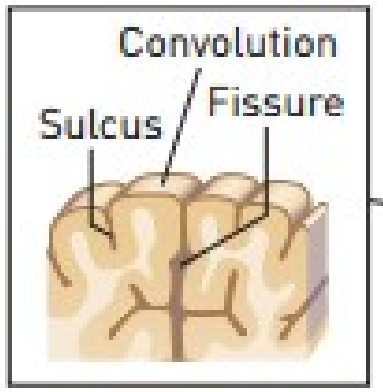
The Cerebrum

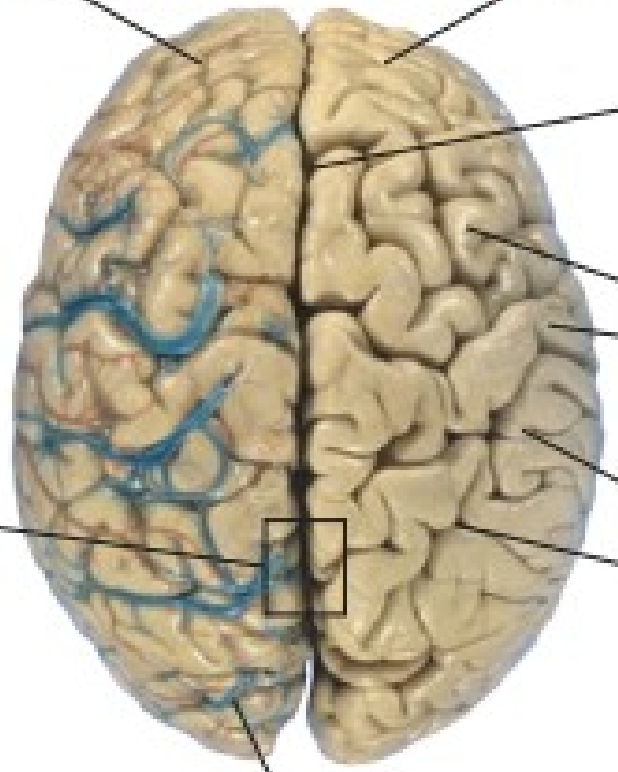
- Folded Surface
  - Increase surface area
  - Fits 70% of neurons in CNS
- Ridges
  - Called Convolutions or Gyri (s. gyrus)
- Downfolds
  - Shallow ones called sulci (s. sulcus)
  - Deep ones called Fissures
- Longitudinal Fissure
  - Separates cerebrum into two hemispheres
  - Corpus callosum joins the hemispheres at the base.



Learning Aim: Describe the main features of the cerebrum

The cerebrum is divided into a **left hemisphere** and a **right hemisphere.**





Longitudinal fissure – a deep cleft that almost separates the cerebrum into two halves, the cerebral hemispheres

Convolutions (or **gyri**) **–** folds of the surface of the cerebrum

Sulci – shallow downfolds of the surface of the cerebrum

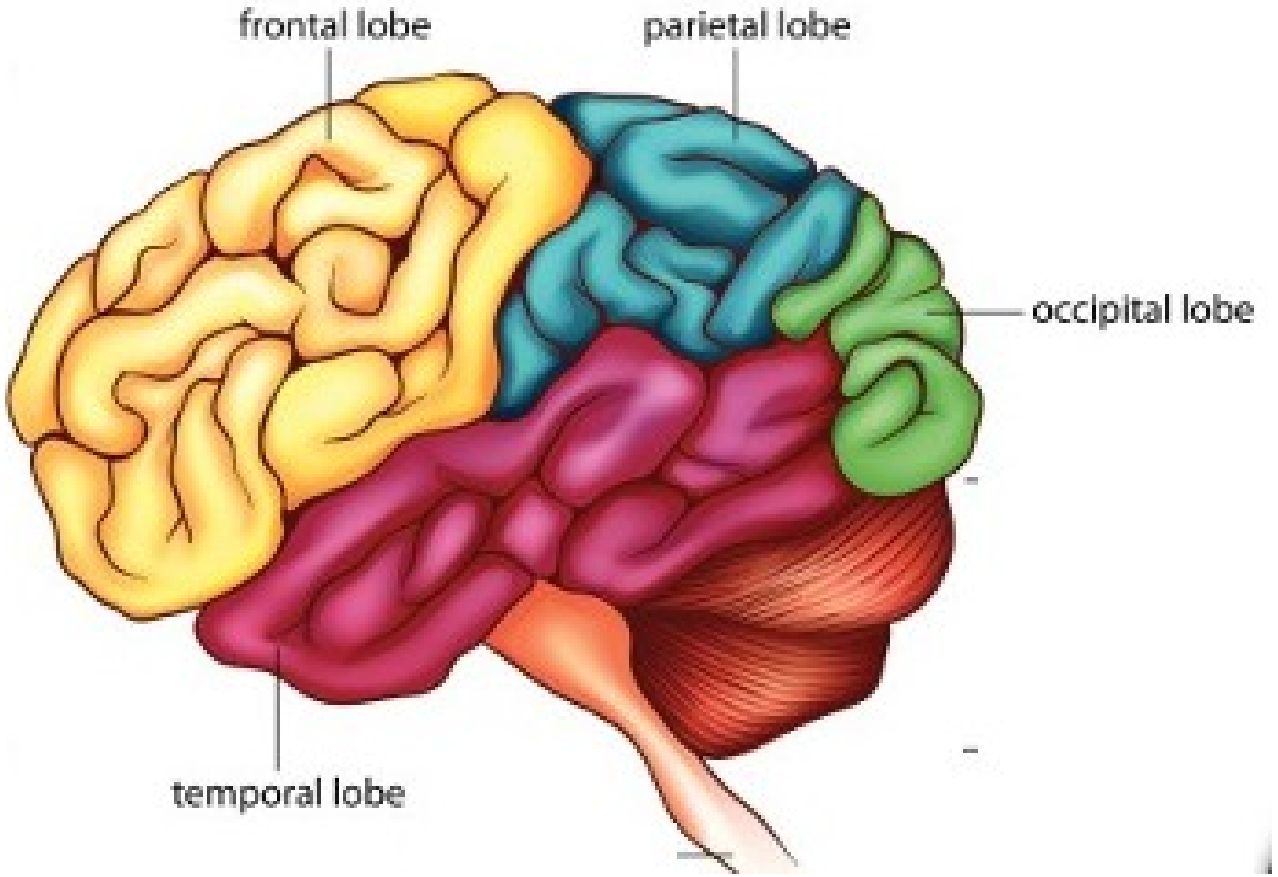
The innermost meningeal layer has been left on this side of the brain so that the blood vessels (coloured blue) can be seen.

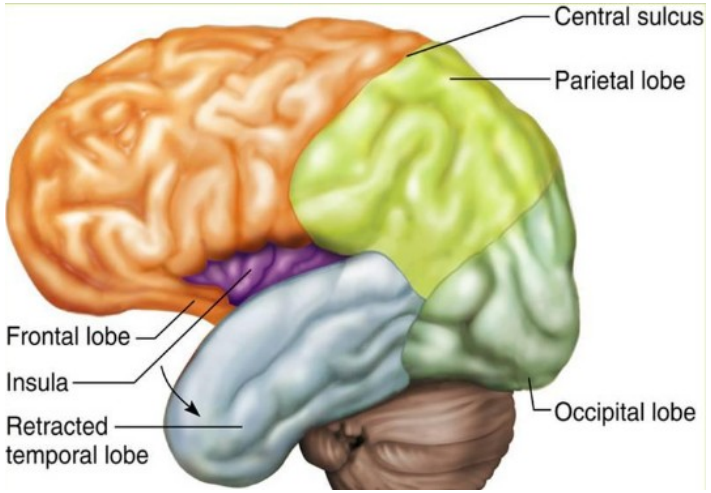
Learning Aim: Describe the main features of the cerebrum

The Cerebrum

- Each hemisphere: 4 lobes
  - Frontal
  - Temporal
  - Occipital
  - Parietal

  - Insula (like a 5th lobe)

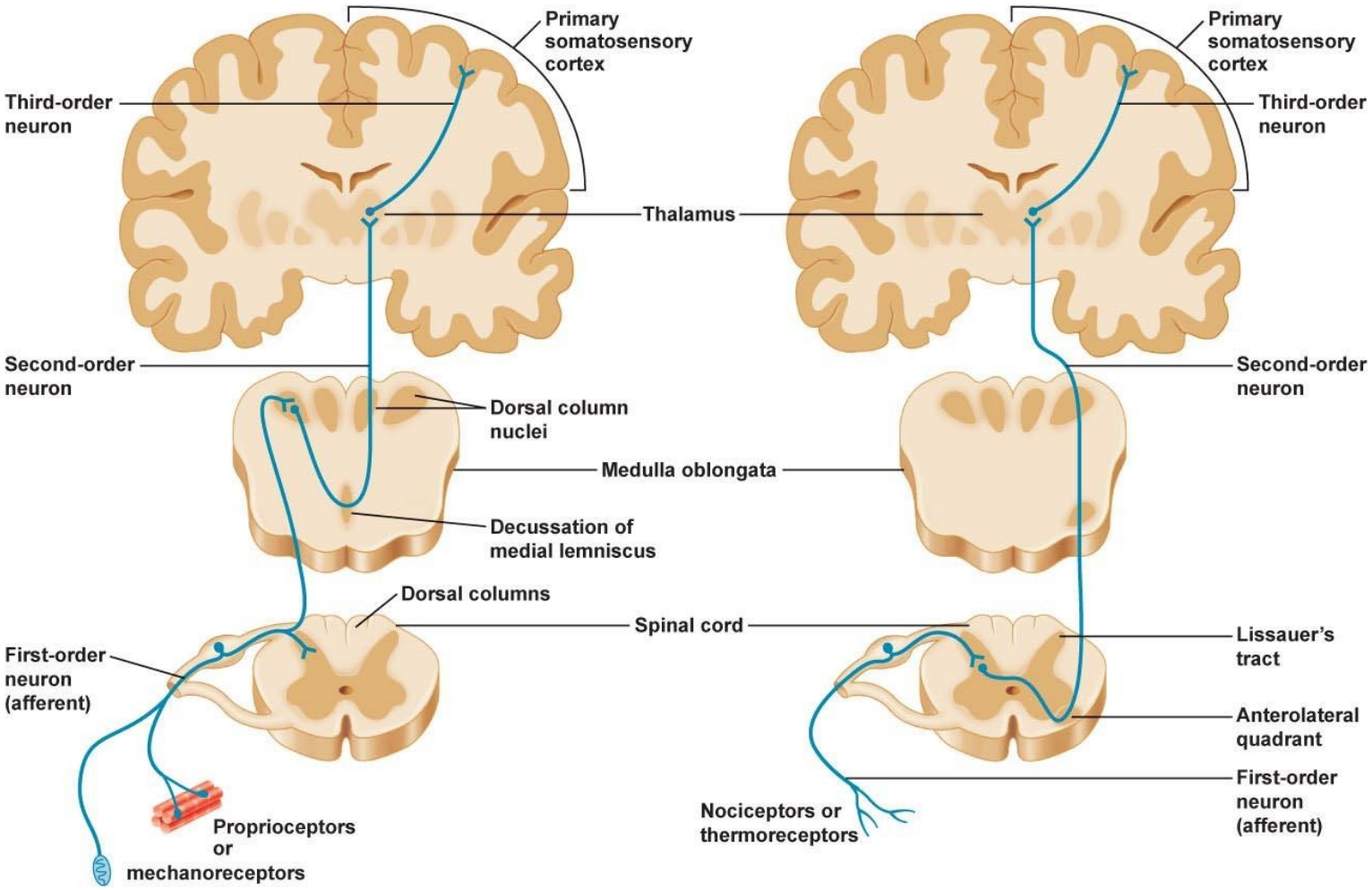




Learning Aim: Describe the main features of the cerebrum

Pathways to the Cerebral Cortex

note: this is a couple of examples only. There are MANY more

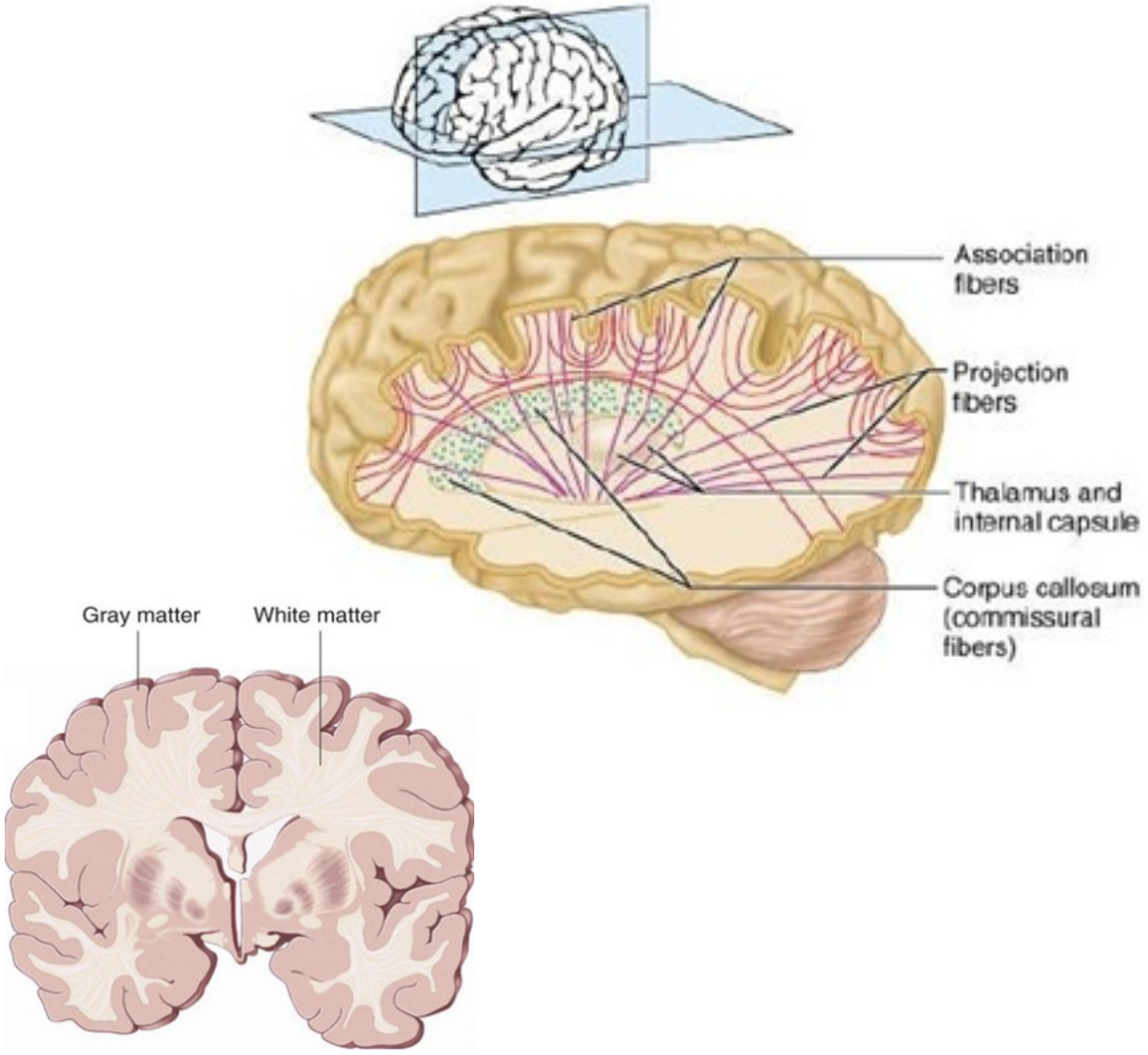


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Learning Aim: Understand some pathways involved in sensory impulses reaching the cerebrum.

Cerebral White Matter

- Myelinated axons
- Bundles of nerve fibres in CNS are called tracts
- 3 types of tracts in white matter:
  - Tracts connecting parts of cortex within same hemisphere – Association Fibres
  - Tracts between left and right hemispheres – Commissural Fibres (including Corpus Callosum)
  - Tracts connecting cortex to other parts of brain or spinal cord – Projection Fibres



Learning Aim: Describe the types of connections (tracts) in the cerebrum

Functions of the Cerebrum

- Thinking, reasoning, learning, memory, intelligence
- Sensory Perception
- Control of voluntary muscle contraction
- Memory held via circuits and links rather than special “cells”

The cerebrum can be studied via:

- Looking at brainwaves: Electroencephalogram
- Changes to blood flow
- Effects of damage and disease
- Experimentation on other mammals.

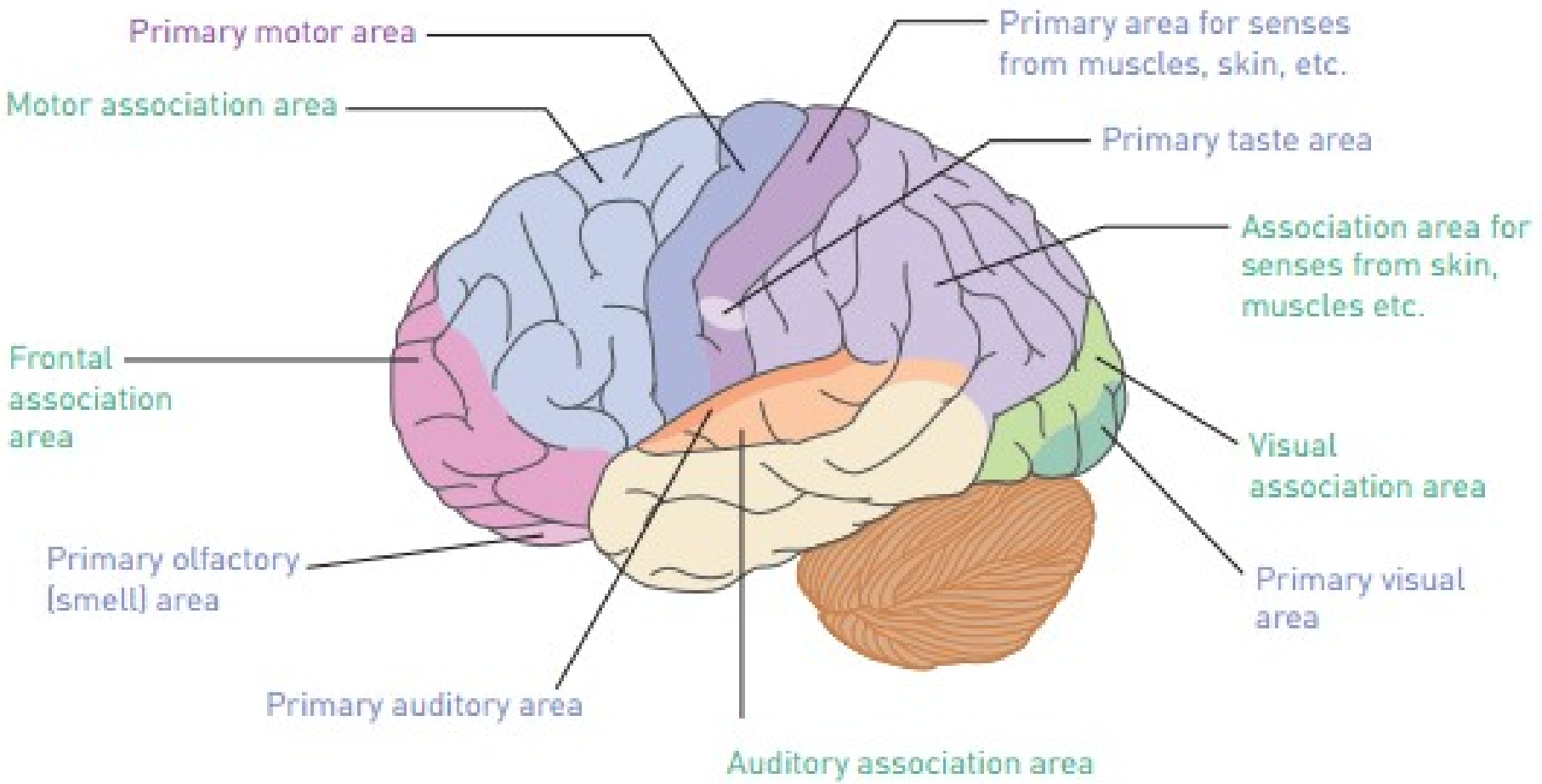
These have enabled the cerebrum to be partially mapped for function

Learning Aim: Describe the functions of the cerebrum and how it can be studied.

Sensory areas receive and process nerve impulses from the senses.

Motor areas send impulses to muscles, especially for voluntary movement.

Association areas interpret information from the senses and make it useful.



Learning Aim: Describe the main areas of the cerebrum and the functions they are responsible for.

Left vs Right Hemispheres – the facts

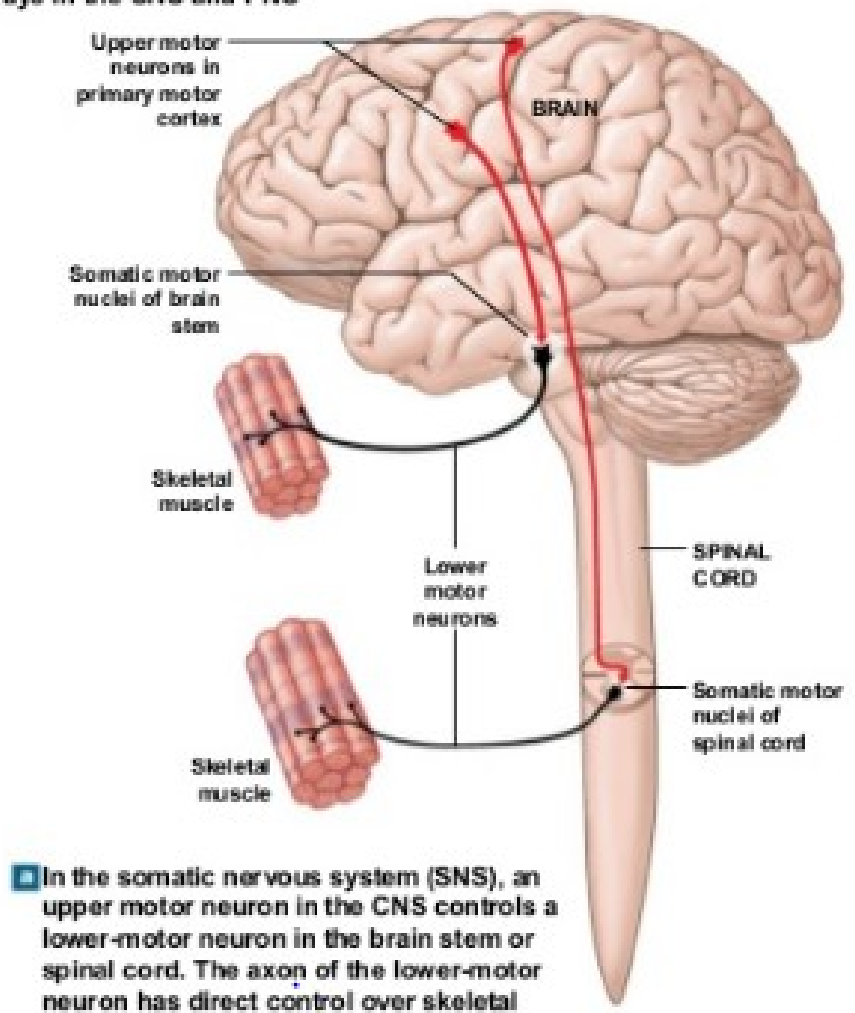
- Some differences eg
  - Language ability – left
  - Music, art – right

- Both sides used equally though.
- People are not “left brained” or “right brained”.

Pathways from the Cerebral Cortex

note: this is one example only. There are MANY more

Figure 15.4a Motor Pathways in the CNS and PNS

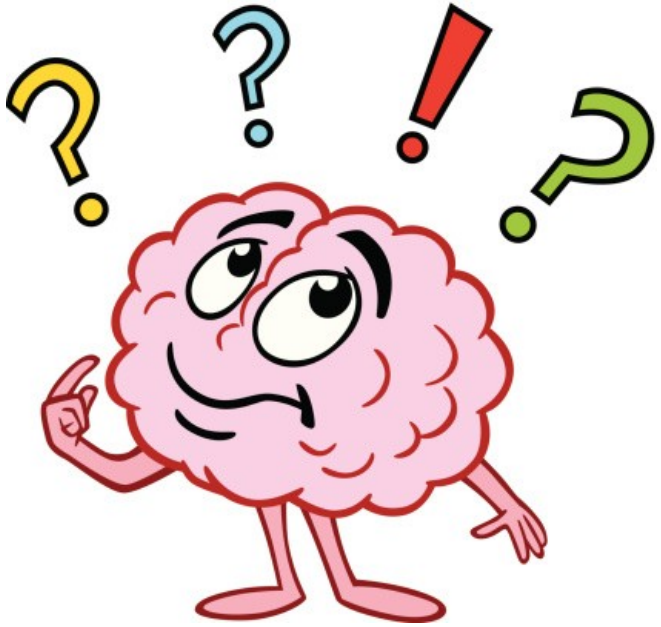


In the somatic nervous system (SNS), an upper motor neuron in the CNS controls a lower-motor neuron in the brain stem or spinal cord. The axon of the lower-motor neuron has direct control over skeletal muscle fibers. Stimulation of the lower-motor neuron always has an excitatory effect on the skeletal muscle fibers.

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Learning Aim: Understand some pathways involved in motor impulses leaving the cerebral cortex.

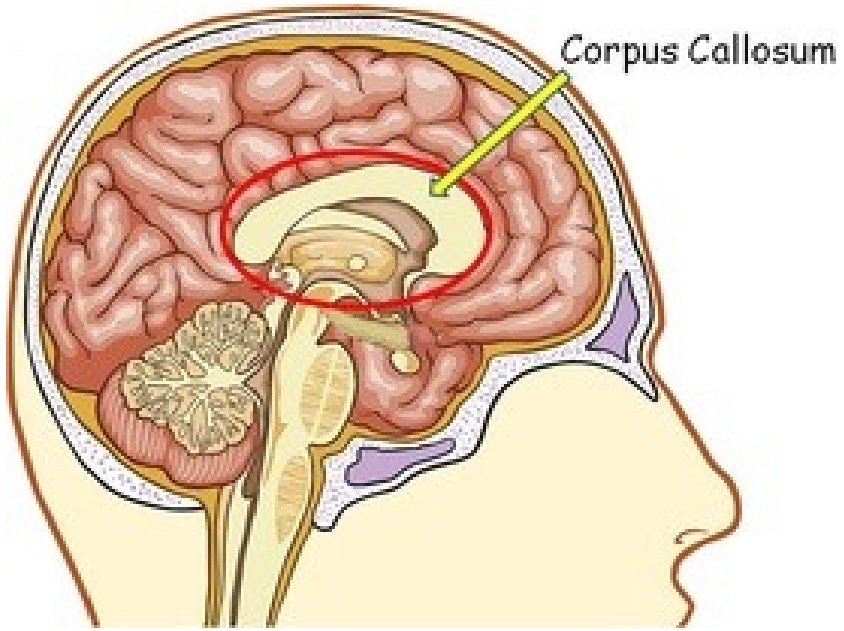
Brain structures other than cerebrum



Corpus Callosum
Cerebellum
Hypothalamus
Medulla Oblongata
Spinal Cord

Corpus Callosum

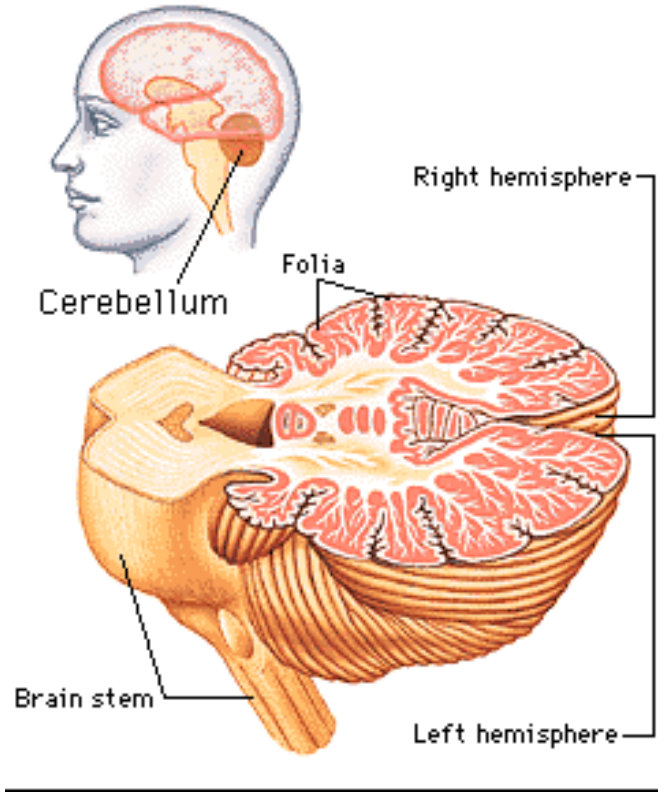
- Wide band of nerve fibres
- At base of longitudinal fissure
- Allow the 2 sides of cerebrum to communicate



Learning Aim: Describe the location and function of the Corpus Callosum.

Cerebellum

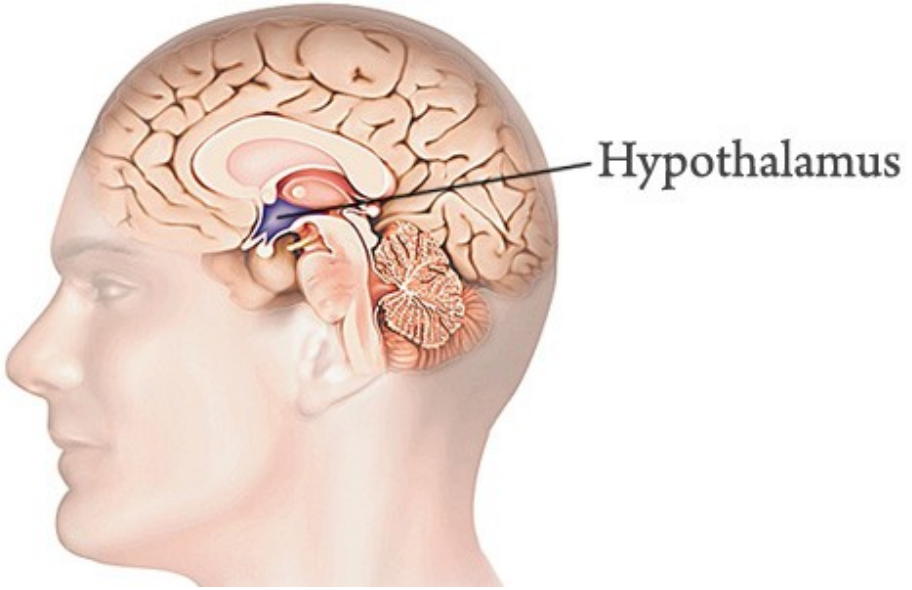
- Under rear of cerebrum
- Highly folded surface
- Control over:
  - Balance
  - Posture
  - Fine coordination of voluntary muscle movement
- Receives input from inner ear, stretch receptors in skeletal muscles.
- “smooths” and coordinates movement.



Learning Aim: Describe the location and function of the Cerebellum.

Hypothalamus

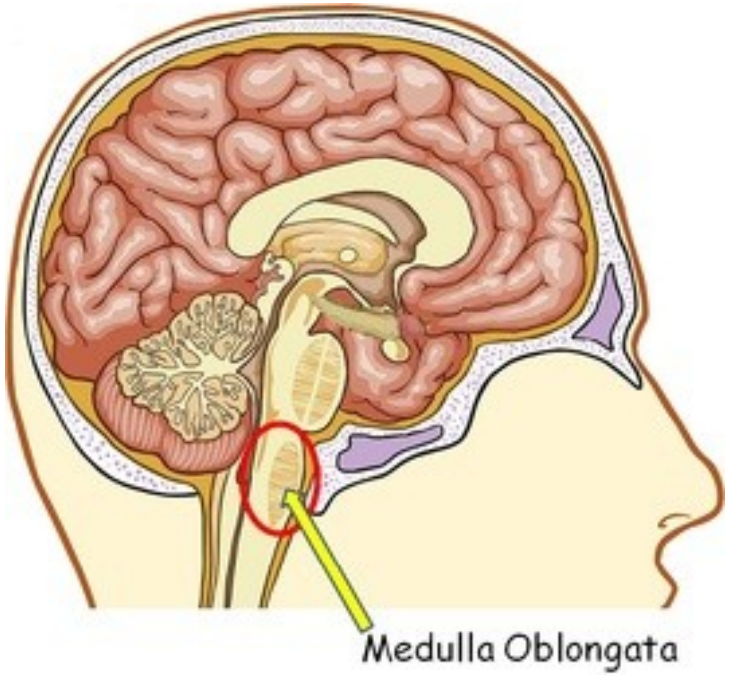
- Deep in middle of brain
- Small but controls many activities
- Homeostatic control
  - Autonomic nervous system coordination
  - Body temperature
  - Food and water intake
  - Emotional responses
  - Coordination of parts of endocrine system
    - Metabolism
    - Growth
    - Reproduction
    - Responses to stress



Learning Aim: Describe the location and functions of the Hypothalamus

Medulla Oblongata

- Continuation of spinal cord at base of brain.
- Nerve fibres pass through on the way to other centres
- Also has role in coordinating key body functions:
  - **Cardiac centre:** rate and force of heartbeat
  - **Respiratory centre:** rate and depth of breathing
  - **Vasomotor centre:** diameter of blood vessels – blood pressure
  - Swallowing, coughing, sneezing, vomiting reflexes

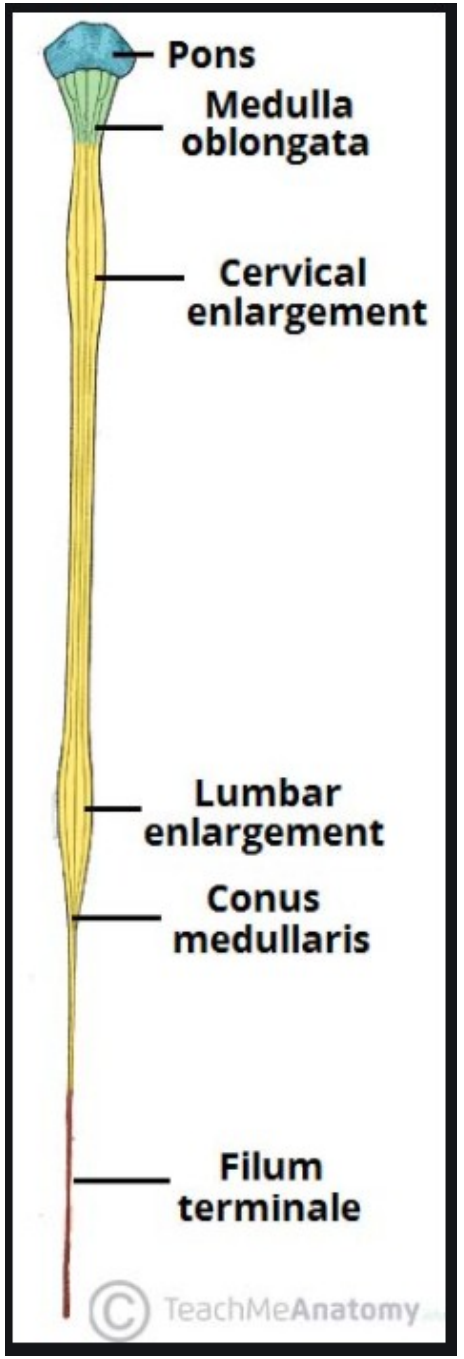


- Interacts with hypothalamus and other higher brain centres to control homeostasis.

Learning Aim: Describe the location and function of the Medulla Oblongata.

Spinal Cord

- Extends from base of skull down spinal column
- Heavily protected by bone, meninges, CSF
- Grey matter at centre, white matter at outside
- White matter in bundles
  - Ascending tracts – sensory axons
  - Descending tracts – motor axons
- Carries impulses up and down between brain and body
- Coordinates some reflexes



Learning Aim: Describe the structure and function of the spinal cord.

Don’t stress – you don’t need to know this in detail – its just to show you where the sensory neurons run and where the motor ones run

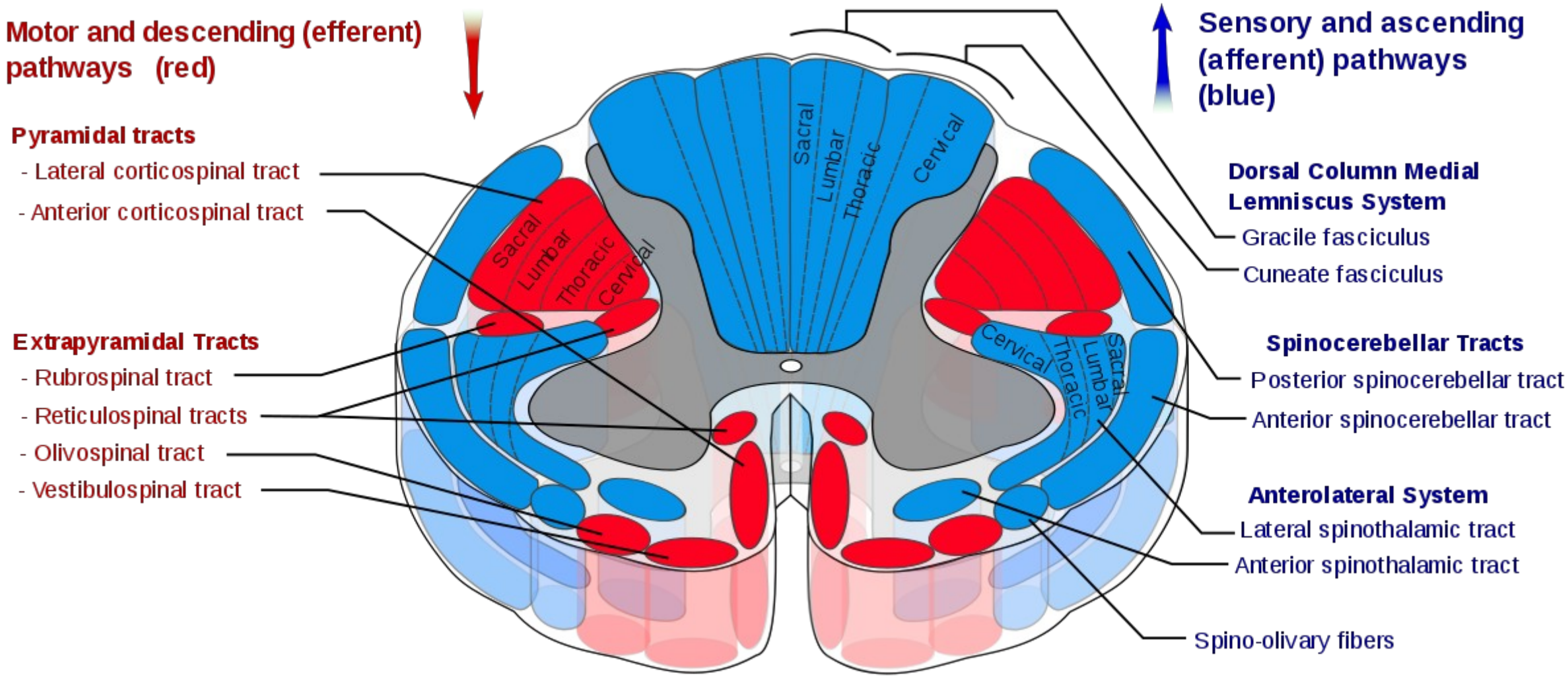


Table 5.1 A summary of the functions of the main functional areas of the CNS

Structure	Function
Cerebral cortex	Higher-order functions such as thinking, reasoning, memory, learning, conscious awareness of surroundings
Corpus callosum	Communication between the two cerebral hemispheres
Cerebellum	Coordination of fine contractions of muscles resulting in smooth movements and the maintenance of posture and balance
Hypothalamus	Homeostasis; regulation of the heart, digestive system, appetite, thirst, metabolism, body temperature, response to fear or anger
Medulla oblongata	Under the influence of the hypothalamus, regulates the heart, breathing and diameter of blood vessels
Spinal cord	Provides a pathway for communication between muscles and glands and the brain. Integration of automatic, protective reflexes